FMLA_Assignment 3

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#A. Create a pivot table for the training data with Online as a column variable, CC as a row variable, and Loan as a secondary row variable. The values inside the table should convey the count. In R use functions melt() and cast(), or function table(). In Python, use panda data frame methods melt() and pivot().

```
# Load the data
df <- read.csv("~/ML/UniversalBank.csv")

# Set seed
set.seed(1)

# Split data into training (60%) and validation (40%)
train_index <- sample(1:nrow(df), 0.60 * nrow(df))

train <- df[train_index, ]
validation <- df[-train_index, ]


#Pivot table: CC (rows), Loan (secondary rows), Online (columns)
pivot_table <- with(train, table(
  CC = CreditCard,
  Loan = Personal.Loan,
  Online = Online
))

ftable(pivot_table)
```

```
##         Online    0    1
## CC Loan
## 0  0            805 1119
##    1             79  119
## 1  0            332  469
##    1             30   47
```

#B. Consider the task of classifying a customer who owns a bank credit card and is actively usingonline banking services. Looking at the pivot table, what is the probability that this customerwill accept the loan offer? [This is the probability of loan acceptance (Loan = 1) conditional onhaving a bank credit card (CC = 1) and being an active user of online banking services (Online= 1)]

```
probability <- with(train,
  mean(Personal.Loan[CreditCard == 1 & Online == 1])
)
```

```r
print(probability)
```

```
## [1] 0.09108527
```

```r
print(probability * 100)  # Percentage
```

```
## [1] 9.108527
```

```r
# C1. Loan (rows) by Online (columns)
table1 <- table(
  Loan = train$Personal.Loan,
  Online = train$Online
)

print(table1)
```

```
##     Online
## Loan    0    1
##    0 1137 1588
##    1  109  166
```

#C. Create two separate pivot tables for the training data. One will have Loan (rows) as a function of Online (columns) and the other will have Loan (rows) as a function of CC.

```r
# Pivot table 1: Loan (rows) as a function of Online (columns)
table1 <- table(
  Loan = train$Personal.Loan,
  Online = train$Online
)

print(table1)
```

```
##     Online
## Loan    0    1
##    0 1137 1588
##    1  109  166
```

```r
# Pivot table 2: Loan (rows) as a function of CC (columns)
table2 <- table(
  Loan = train$Personal.Loan,
  CC = train$CreditCard
)

print(table2)
```

```
##     CC
## Loan    0    1
##    0 1924  801
##    1  198   77
```

#D. Compute the following quantities [P(A | B) means "the probability ofA given B"]:i. P(CC = 1 | Loan = 1) (the proportion of credit card holders among the loan acceptors)ii. P(Online = 1 | Loan = 1)iii. P(Loan = 1) (the proportion of loan acceptors)iv. P(CC = 1 | Loan = 0)v. P(Online = 1 | Loan = 0)vi. P(Loan = 0

```
# Convert variables to numeric if they are factors or characters
train$CreditCard <- as.numeric(as.character(train$CreditCard))
train$Online <- as.numeric(as.character(train$Online))
train$Personal.Loan <- as.numeric(as.character(train$Personal.Loan))

# Calculating the six probabilities
p1 <- mean(train$CreditCard[train$Personal.Loan == 1])
p2 <- mean(train$Online[train$Personal.Loan == 1])
p3 <- mean(train$Personal.Loan == 1)

p4 <- mean(train$CreditCard[train$Personal.Loan == 0])
p5 <- mean(train$Online[train$Personal.Loan == 0])
p6 <- mean(train$Personal.Loan == 0)

# Results
p1
```

```
## [1] 0.28
```

```
p2
```

```
## [1] 0.6036364
```

```
p3
```

```
## [1] 0.09166667
```

```
p4
```

```
## [1] 0.293945
```

```
p5
```

```
## [1] 0.5827523
```

```
p6
```

```
## [1] 0.9083333
```

#E. Use the quantities computed above to compute the naive Bayes probability P(Loan = 1 | CC= 1, Online = 1)

```r
# Numerator: Loan = 1
num <- p1 * p2 * p3

# Denominator: Loan = 1 plus Loan = 0
den <- (p1 * p2 * p3) + (p4 * p5 * p6)

# Naive Bayes probability
prob <- num / den

# Display probability and percentage
prob
```

```
## [1] 0.09055758
```

```r
prob * 100
```

```
## [1] 9.055758
```

The Naive Bayes probalility is 0.090, it shows that only 9% chance of accepting the loan.The overall acceptance rate in the data is very low, so neither credit card nor online strongly differentiate acceptors from non - acceptors.

#F. Compare this value with the one obtained from the pivot table in (B). Which is a more accurate estimate?

The pivot table result from Part B is more accurate for the training data because it is calculated directly from the actual customer records. The Naive Bayes result from Part E is based on the assumption that credit card ownership and online banking use are independent given the loan outcome. If this assumption is not true, the estimate may be less accurate. Hencce, the pivot table gives the actual probability for the training data, while Naive Bayes estimates the probability using this assumption.

G. Run Naive Bayes

```r
# install.packages("e1071")
library(e1071)

# Convert variables to factors
train$Personal.Loan <- as.factor(train$Personal.Loan)
train$CreditCard <- as.factor(train$CreditCard)
train$Online <- as.factor(train$Online)

# Build the Naive Bayes model
nb_model <- naiveBayes(
  Personal.Loan ~ CreditCard + Online,
  data = train
)
```

```r
# Create a customer with CC = 1 and Online = 1
new_customer <- data.frame(
  CreditCard = factor("1", levels = levels(train$CreditCard)),
  Online = factor("1", levels = levels(train$Online))
)

# Get predicted probabilities
nb_result <- predict(
  nb_model,
  newdata = new_customer,
  type = "raw"
)

# Display the probabilities
print(nb_result)
```

```
##              0          1
## [1,] 0.9094424 0.09055758
```

```r
# Extract probability for Loan = 1
nb_probability <- nb_result[1, "1"]
print(nb_probability)
```

```
##          1
## 0.09055758
```

```r
# Compare with Part E
print(prob)
```

```
## [1] 0.09055758
```

```r
print(nb_probability - prob)
```

```
## 1
## 0
```

#The probability found in Part B shows the actual loan acceptance rate among training customers who use online banking and have a credit card whereas the Naive Bayes method in Part E estimates the probability by assuming that the two predictors are conditionally independent given the loan outcome. The result in Part E was approximately 9%. This comparison shows how Naive Bayes uses customer information to predict the likelihood of accepting a personal loan.